

How do state fiscal and regulatory policies impact their economic performance? Evidence from 2000-2020.

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I. Introduction

Understanding how public policies impact economic output is of immense importance to lawmakers, researchers, and the lay voter. The United States is a federation, meaning that it has a federal government and state governments that also have independent taxation, spending, and lawmaking power. This ability for independent innovation and implementation of public policy ideas allows for one to observe these for their economic impacts, this “laboratory of democracy” is invaluable for data collection and research on a host of public finance questions.

To what extent, if any, does the level of taxation and regulation—differing measures of government largess—impact economic output both overall and for private sector activity? To what extent, if any, do high public sector spending and burdensome regulations reduce the market’s ability to produce and lead to inefficiencies that may otherwise not exist had a laissez-faire approach been adopted? As some American states have some but not all these traits, researchers can make empirical assessments of these kinds of questions by running sophisticated econometric models on collected information.

II. Data Used

As this project seeks to understand the effect of state policies, a large data set of information with these details must be used for these variables are the independent variables used to predict state economic output and that output attributable to the private sector. The CATO Institute’s Freedom in the 50 States report was invaluable to this end. It has data on approximately 200 variables relating to state government fiscal, regulatory, and personal freedom policies. Although personal freedom policies were outside this project's scope, this project was able to utilize a limited sample of 19 of the fiscal and regulatory policies that were

tracked in the CATO spreadsheet. Subsequent paragraphs will provide explanations for the variables.

Many variables of interest are aspects of fiscal policy in proportion to the size of a given state's economy. State government consumption and investment looks at the total spending by a state. Local tax and state tax look at the tax burden imposed by municipalities and states, respectively. The log of competing jurisdictions measures the weighted sum of local governmental entities in a state; these weights are based on the revenue collected by each jurisdiction divided by the square miles of private land in a state. Debt as a percentage of state economic output and the total size of state and local governmental payrolls relative to the size of private sector payrolls are included as variables. The sum of fiscal policy considers state tax revenues, government consumption, local tax revenues, employment, taxation and state cash and security assets. The sum of local taxation is the amount of taxation (relative to state incomes) carried out by local governmental entities (counties, villages, etc.), considering the number of competing jurisdictions; the reasoning behind including this is that a state with more competing jurisdictions affords more of an opportunity for its residents to relocate to lower-tax jurisdictions, lessening the impact of local taxation. Some of the CATO variables, such as sum local tax will consider variables together rather than on a stand-alone basis.

Regulatory compensation and assessment required asks whether a state has an independent body to provide compensation and prevent regulatory overreach by state government. This is a dummy variable with 2 meaning that both (assessment/compensation) is required, 1 means that only 1 (assessment *or* compensation) is required, these dummies are multiplied by .5 if these rules only apply to state agencies, not localities. Eminent domain reform is a dummy variable with 0 meaning no reform and 1 with reforms that limit the power of state government to legally compel property owners to sell. 1 is multiplied by 1.5 if the protections are

in a state's constitution and are stricter than federal constitutional standards. The Wharton land use regulatory index is composed of 11 sub-indexes. 9 of these relate to local rules and 2 relate to a state's laws and court system. A higher index number means more restrictions are in place. The minimum to median wage ratio takes the minimum hourly wage for each state and divides it by the median hourly wage for that state, which lets the variable consider differing labor market conditions when assessing minimum wage policies. Discrimination law is a dummy variable for if state law covers a protected class not covered by the federal government with 1 meaning yes and 0 meaning no. Sum health insurance freedom considers state regulations on health insurance ranging from community risk v. individual risk rules, insurance mandates and mandated coverages. A lower score means more restrictions and less deregulation has occurred. Sum telecom freedom measures if a state has enacted policies such as allowing entrant firms access to use local telephone and TV cable lines, among other reforms. A higher score means more deregulations have been enacted. The sum of education and experience requirements is the sum of every individual regulation requiring workers to have a specific level of education/experience to work in a specific occupation. Other regulation freedom looks certificate-of-need requirements for new hospital construction, auto insurance rate filing requirements, homeowner's insurance rate filing requirements, among others.

This project uses two measures for economic output when reporting results from two regressions; real (inflation adjusted) GSP (gross state product) and real GSP per capita attributable to the private sector. The total size of each state's economy and that of its private sector was sourced from the US BEA (Bureau of Economic Analysis) regional analysis project downloadable data files. I then accessed data files from the census which had the decadal populations and yearly population estimates from every state. Taking both real GSP and real

private GSP from BEA and dividing its population gets GSP in terms of population—per capita GSP and per capita private GSP.

Table 1 shows summary statistics from these variables

Table 1 summary statistics

Variable	Mean	Std.	Min	Max
Stategovconsumptioninvestmentdi	0.108	0.017	0.067	0.177
localtaxpercentincome	0.039	0.010	0.016	0.087
logcompetingjursdiction	0.707	0.490	0.074	1.998
statetaxpercentincome	0.058	0.015	0.011	0.122
debtpercentincome	0.187	0.052	0.048	0.437
statelocalemploymentdivrivatetem	0.130	0.023	0.078	0.211
sumfiscal	0.006	0.150	-0.533	0.487
sumlocaltax	0.002	0.071	-0.269	0.192
regcompensationassessmentrequire	0.479	0.594	0	2
eminentdomainreform	1.667	1.326	0	4.5
Whartonlanduseregulationindex	-0.036	0.638	-1.130	2.32
minimumtomedianwageratio	0.418	0.057	0.257	0.655
discriminationlaw	0.875	0.331	0.000	1
sumhealthinsurancefreedom	0.000	0.066	-0.087	0.122
sumcabletelecomm	0.477	0.500	-0.004	1.005
sumeducexperrequirements	20633.650	15166.080	5976	66254
sumoccupationalfreedom	0.000	0.014	-0.036	0.036
otherregulationfreedom	0.000	0.012	-0.026	0.030
nonfederalregulatorysum	0.000	0.125	-0.386	0.190
n=1150				

State government consumption and investment and local tax as a percentage of income have a standard deviation of 0.017 and 0.010, suggesting there is limited variation in some dimensions of fiscal policy across states. More variation in the log of competing local jurisdictions can be observed (standard deviation of 0.490), which is why the sum of local tax has a higher standard deviation 0.071 than local tax percent of income 0.010.

Regulatory policy variables appear to vary more among states than fiscal policy variables. Eminent domain reform and Wharton land use regulatory index have a standard deviation of 1.326 and 0.638, respectively. Significant differences in the degree of regulation over the telecom industry and anti-discrimination law coverage can also be observed with a standard deviation of 0.500 and 0.331, respectively.

The two equations to be estimated take the following forms of:

$$\text{Per capita GSP} = \beta_0 + \beta_1 \text{state gov consumption investment di} + \beta_2 \text{local tax percent income} + \beta_3 \log \text{compet} \\ \text{ing jurisdiction} + \beta_4 \text{state tax percent income} + \beta_5 \text{debt percent income} + \beta_6 \text{state local employment div riva} \\ \text{teem} + \beta_7 \text{sum fiscal} + \beta_8 \text{sum local tax} + \beta_9 \text{reg compensation assessment require} + \beta_{10} \text{eminent domain re} \\ \text{form} + \beta_{11} \text{wharton land use regulation index} + \beta_{12} \text{minimum to median wage ratio} + \beta_{13} \text{discrimination l} \\ \text{aw} + \beta_{14} \text{sum health insurance freedom} + \beta_{15} \text{sum cable telecom} + \beta_{16} \text{sum educ exper requirements} + \beta_{17} \text{sum occupational freedom} + \beta_{18} \text{other regulation freedom} + \beta_{19} \text{non federal regulatory sum} + u$$

And...

$$\text{Per capita private GSP} = \beta_0 + \beta_1 \text{state gov consumption investment di} + \beta_2 \text{local tax percent income} + \beta_3 \log \\ \text{competing jurisdiction} + \beta_4 \text{state tax percent income} + \beta_5 \text{debt percent income} + \beta_6 \text{state local employment} \\ \text{div rivateem} + \beta_7 \text{sum fiscal} + \beta_8 \text{sum local tax} + \beta_9 \text{reg compensation assessment require} + \beta_{10} \text{eminent do} \\ \text{main reform} + \beta_{11} \text{wharton land use regulation index} + \beta_{12} \text{minimum to median wage ratio} + \beta_{13} \text{discrimi} \\ \text{nation law} + \beta_{14} \text{sum health insurance freedom} + \beta_{15} \text{sum cable telecom} + \beta_{16} \text{sum educ exper requirem} \\ \text{ents} + \beta_{17} \text{sum occupational freedom} + \beta_{18} \text{other regulation freedom} + \beta_{19} \text{non federal regulatory sum} + u$$

III. Results

Table 2 depicts the results of the fixed effects regression for the panel data. Fixed effects is most appropriate for state data as various states seem to consistently under or over perform economically relative to others for reasons that are exogenous to this model. For example, West Virginia's underwhelming economic performance is the result of the declining coal industry, which this model cannot account for directly. Other states may overperform as well, such as Florida due to favorable climate conditions that draw in retirees whose spending boosts gsp above what may otherwise occur.

I found that every percentage point increase in state government spending reduces both per capita gsp and private sector gsp by more than \$200 per person. Every percentage point increase in state tax burden reduces per capita gsp and per capita private gsp by more than \$500 and \$600, respectively. Sum of fiscal policy variables shows an overall one percentage point increase in tax burden by both state and local governments when accounting for competing jurisdictions reduces per capita gsp and per capita private gsp by more than \$50 and \$60 per person, respectively. Not only are these results practically significant, but they are also statistically significant.

Not all regressed variables are significant in either a statistical or practical sense. The given numbers are for per capita gsp, the results for private per capita gsp are similar. State debt burden relative to size of economy, -2.868, size of public sector payrolls relative to private sector payrolls -20.37, regulatory assessment required 0.0769, minimum-to-median wage ratio 0.269, telecom deregulation -0.357 and miscellaneous regulatory policies 10.82, lack both practical and statistical significance. Other variables are statistically significant, but not practically significant. These include eminent domain reform -0.394, Wharton land use regulatory index 0.502, discrimination law -1.54, health care market deregulation -3.630 and telecom industry deregulation -0.357.

Table 2 provides a summary of the co-efficients and standard errors of the two fixed-effects regressions that were run for this project.

Table 2 Regression Results

(1)	(2)
percapitagsp	percapitaprivategsp

State government consumption & income divided by income	-230.6*	-265.7*
	(99.75)	(128.3)
Local taxation as a percent of income	267.7***	343.0***
	(74.13)	(95.36)
Log of competing local jurisdictions in a state	8.443***	10.44***
	(2.190)	(2.817)
State taxation as a percent of income	-500.4**	-617.0**
	(175.2)	(225.3)
State debt as a percent of income	-2.868	-3.548
	(2.189)	(2.816)
State & local payrolls divided by private sector payrolls	-20.37	-21.28
	(20.92)	(26.92)
Sum fiscal policy variables	-54.83**	-67.51*
	(21.19)	(27.25)
Sum local tax variables	97.42***	123.3***
	(22.27)	(28.65)
Regulatory assessment required	0.0769	-0.0272
	(0.202)	(0.259)
Eminent Domain Reform	-0.394***	-0.525***
	(0.0435)	(0.0559)

Wharton land use regulation index	0.502** (0.173)	0.580** (0.223)
Minimum-to-median wage ratio	0.269 (1.089)	0.405 (1.401)
Discrimination statues cover protected class not under federal law	-1.554*** (0.345)	-2.386*** (0.444)
Degree of deregulation in the health insurance market	3.630*** (0.797)	5.524*** (1.025)
Degree of deregulation of telecom sector	-0.357 (0.287)	-0.457 (0.369)
State education and experience requirements	0.0000535 (0.0000334)	0.0000591 (0.0000429)
Sum of occupational freedom	36.56*** (8.077)	46.84*** (10.39)
Miscellaneous regulatory policies	10.82 (11.07)	9.276 (14.24)
Sum of state regulations	-2.135 (2.274)	-3.305 (2.926)
_cons	62.33* (24.23)	72.98* (31.17)

N	1050	1050
R^2	0.452	0.513

Standard errors in parentheses
 * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

IV. Commentary and Conclusions

Some outputs from this regression were surprising such as the minimum wage and telecom industry not being significant. The non-impact of the minimum wage in this study may be because low-skilled workers who are dislocated because of minimum wage increases make up a small share of the economy. The non-impact of telecom regulations in this study is likely because telecom is a minor economic sector in its own right; the impact on consumers of telecom deregulation (better industry performance, i.e., lower prices on phone bills) may not be material, meaning that few people or companies are going to move to a state for the sole reason of getting a cheaper phone bill; also cable TV impact is a factor, but given that many people in contemporary society have eschewed cable, this is of no relevance to them. Both non-impacts could also be occurring as this study only examined 19 policy variables, a more robust study with more variables may arrive at different co-efficient estimates for these variables. It is also important for policy makers to remember that different metrics of economic performance should also be considered before making decisions. Even if a more robust analysis found that per capita gsp is not impacted by minimum wage increases, dislocation of low-skilled workers from the labor force may occur if the wage is set above the market clearing wage. Policy makers may wish to avoid higher unemployment, even if causing higher unemployment would not cause gsp impact, whether such an impact would occur though is beyond the scope of this paper.

Overall, it appears that fiscal policy is more impactful on state economic performance than regulatory policy. The fiscal policy co-efficients were significantly larger than the regulatory policy co-efficients. Also, many regulatory policies do not appear statistically significant, this is not the case with the fiscal policy variables.

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